

Integrating pulmonary surfactant into lung mechanical simulations: A continuum approach to surface tension in poromechanics

Nibaldo Avilés-Rojas^{a,b}, Daniel E. Hurtado^{a,b,c,1}

^a*Department of Structural and Geotechnical Engineering, School of Engineering, Pontificia Universidad Católica de Chile, Santiago, Chile, Vicuña Mackenna 4860, Santiago, Chile*

^b*Institute for Biological and Medical Engineering, Schools of Engineering, Medicine and Biological Sciences, Pontificia Universidad Católica de Chile, Vicuña Mackenna 4860, Santiago, Chile*

^c*Institute for Medical Engineering and Science, Massachusetts Institute of Technology, Cambridge, MA, 02140, USA*

Abstract

Surface tension arising in the air-liquid interface of alveolar tissue is a fundamental mechanism in lung physiology that explains lung recoil and hysteresis during breathing. Despite its importance, surface tension dynamics are poorly addressed in computational models of the lungs, due to its complex multiscale physicochemical nature. In this study, we formulate a poromechanical framework that seamlessly incorporates the effect of surfactant-dependent surface tension in porous media for the prediction of lung hysteretic response. Using an internal variable formalism, we apply the Coleman-Noll procedure to establish an expression for the stress tensor that includes surface tension akin to the Young-Laplace law. Based on this formulation, we construct non-linear finite-element models of human lungs to simulate pressure-volume curves and lung response during mechanical ventilation. Our results show that surfactant-dependent surface tension notably modulates pressure-volume curves and lung mechanics. In particular, our model captures the influence of surfactant dynamics on lung hysteresis and compliance, predicting the transition from an insoluble reversible regime to a dissipative one governed by Langmuir kinetics. We envision that our continuum framework will enable lung simulations where surfactant-related phenomena are directly considered in predictions, with important applications to modeling respiratory disease and lung response to mechanical ventilation.

Keywords: Surface tension, Lung modeling, Pressure-volume curve, Internal variables, Pulmonary surfactant

1. Introduction

The lungs are highly porous organs whose mechanical response is fundamental to sustain life. During breathing, the lung mechanical function is largely driven by the air-liquid-tissue

^{*}Corresponding Author: Daniel E. Hurtado, email address: daniel.hurtado@uc.cl, Phone: +562 2354 4207

interaction arising in alveoli, the main microstructural components of the lung parenchyma. Parenchyma deformation is mediated by the interaction of forces in the alveolar walls and the surface tension that results from the air-liquid interface present in the alveolar cavity. This interfacial phenomenon plays a crucial role in respiratory function mainly due to the large pulmonary surface area-to-volume ratio: approximately 85 m^2 of alveolar surface area are contained in a lung whose volume is roughly 4 liters [80]. In effect, surface forces in the air-liquid interface may account for up to 50% of the elastic recoil in human lungs [33, 56]. One of the most notable features of pulmonary surface tension is its dynamic behavior, where extremely low values are attained to facilitate the breathing process. Such dynamic evolution is explained by the physicochemical activity of the pulmonary surfactant, a complex mixture of proteins and lipids that is adsorbed onto, desorbed from, and squeezed out of the air-liquid interface [35, 58]. In contrast to clean air-water interfaces, where surface tension has values near 70 dyn/cm , alveolar surface tension varies dynamically during the respiratory cycle, reaching values near 0 dyn/cm during the expiratory phase, avoiding alveolar collapse and allowing normal breathing [66, 77]. Further, alterations in the metabolism and composition of the surfactant are related to pathological conditions, such as acute respiratory distress syndrome where the abnormally high levels of surface tension cause alveolar collapse, which impedes gas transfer and pulmonary function [3, 24, 63].

The contribution of alveolar surface tension to lung mechanics has received less attention than the effect of alveolar wall elasticity on lung compliance, possibly due to its complex multiscale behavior, which depends on the physicochemical nature of surfactant activity and its hysteretic nature. Traditionally, the lung response has been analyzed using compartment models, in which elastic balloons are connected to resistive conduits, allowing a simplified representation of the interaction between the lung parenchyma deformation and the air flowing through the airways [7, 12, 55, 84]. In these models, the surface tension has been represented by increasing lung compliance, a parameter that represents the ability of the lung to elastically deform and can be estimated experimentally as the slope of the pressure-volume curves [8, 51]. To address the spatial distribution of alveolar pressure regional airflow in the lungs, compartment models have been extended toward lumped-like fractal models, in which an image-based reconstructed structural airway tree is connected with elastic spherical acinar units whose mechanical response follows empirical equations estimated from P-V curves [78]. Broadly, these models incorporate the contribution of surface tension by stiffening the acinar response by adjusting the parameters of the P-V curves [53] or by adding a collapsing pressure induced by the surface tension (Laplace Law) directly into the equilibrium of the acinar transmural pressure [47]. Although the latter approach represents the surface tension contribution in the organ response, it lacks kinematic compatibility as discrete acinar units are typically isolated and do not interact with neighboring tissue nor allow for inter-acinar flow.

Poromechanics theory offers a continuum framework that, by construction, adequately captures the interaction arising in neighboring alveolar units and their deformation caused by airflow. Large-strain poroelastic lung models have been formulated to study the 3D distri-

bution of deformation and pressure in idealized dynamic and quasi-static conditions [9, 59]. More recently, finite-element simulations based on poromechanical formulations have successfully captured critical physiological variables during mechanical ventilation with clinical relevance [4]. Our group also proposed a poroelastic multi-scale finite-deformation framework that takes into account the influence of tissue microstructural properties to predict whole-lung elastic response [16, 17, 29]. Although these poroelastic models offer valuable insights into respiratory mechanics with spatiotemporal resolution, they lack capturing the hysteresis observed during the respiratory cycle, which can be largely attributed to alveolar surface tension. In the last decades, this classical poromechanics framework has been extended to consider elastic surface effects in the interface between the solid matrix and the fluid phases, being applied mainly to the study of differential swelling induced by the adsorption of carbon dioxide in coals during methane production [76, 82]. One of the critical steps in this formulation is the extension of the Helmholtz skeleton free energy potential, adding a term related to the surface energy stored in the matrix-fluid interface. This energy implies an additional stress contribution resulting from the Coleman-Noll procedure, which also considers the geometry of porous media to describe the interface. In addition, surface forces at the interface depend on the adsorption process, resulting in a stress tensor that depends on the concentration of the adsorbed compound. These models constitute a starting point to obtain a representative expression of alveolar stress in a thermodynamically consistent way. However, the applicability of this approach to modeling biological tissues, and the lung parenchyma in particular, poses the challenge of incorporating hysteretical behavior that results from internal non-equilibrium processes such as the dynamic adsorption and desorption of pulmonary surfactant, which has not been addressed to date.

In this work, we propose a continuum framework for lung parenchyma mechanics that seamlessly incorporates the surfactant-dependent surface tension in alveoli, allowing us to predict the hysteretic response of lungs. To this end, we formulate a continuum poromechanical framework with surface effects in which the air-liquid interface surfactant concentration directly modulates the surface tension. A novel aspect of this formulation is the consideration of the surface tension in continuum whole lung mechanics modeling. This paper is organized as follows. In Section 2, we formulate our lung model, starting with the balance laws and thermodynamic restrictions governing porous media mechanics. Next, considering an internal variable framework to track the surfactant mass evolution, we apply the Coleman-Noll procedure to find a general constitutive relationship for the stress tensor representing the lung parenchyma response. This relationship is particularly developed for pulmonary alveoli, represented by a spherical shell geometry and a known surfactant model. Strong and weak formulations of this problem are also introduced. Non-linear finite-element discretizations of the weak formulation are further constructed to simulate pressure-volume curves during supersyringe tests and mechanical ventilation therapy, whose details and results are presented in Section 3. Finally, in Section 4, we discuss our model formulation and its advantages over state-of-the-art virtual lungs.

2. Modeling of lung mechanics with alveolar surface tension effects

To model the effect of surface tension in the lung, we adopt a multiscale approach where each point in the continuum domain will contain interacting phases at the microstructural level, see Figure 1 for a schematic. The lung parenchyma domain is composed of the alveolar structure and the alveolar airspace. The alveolar structure is partitioned into the connective tissue and the liquid lining layer that covers the alveolar walls, see inset in Figure 1. Further, we assume that the liquid lining layer contains surfactant in its bulk solution and on the airspace interface. Moreover, the alveolar airspace is assumed to be fully filled with air that flows throughout the lung domain.

Due to its physicochemical behavior, the surfactant can undergo dynamic processes of adsorption and desorption within the liquid lining layer subsystem, i.e., between its bulk solution and its interface with the air. For simplicity, we will assume that the liquid lining layer in one alveolar unit is not connected with other alveolar units, so that surfactant cannot be transported across the lung parenchyma. Under this assumption, the lining liquid layer can be assumed to be attached to the alveolar tissue, allowing us to consider them as a single phase. Based on these definitions, the lung parenchyma can be described as a two-phase saturated porous medium with air and structural phases. We note that the surfactant dynamics within the alveolar structure will be represented as an internal process governed by thermodynamic considerations, as described below.

2.1. Kinematics and balance laws in the lung parenchyma

Considering a continuum framework [18], we assume that an infinitesimal element of parenchyma and air coincide at a macroscopic current point $\mathbf{x}$, see Figure 1. The Lagrangian observer follows the deformation of the parenchyma and not of the air, so that the current (Eulerian) configuration at a time t is defined as $\Omega = \boldsymbol{\varphi}(\Omega_R, t)$, where Ω_R is the parenchyma reference (Lagrangian) configuration and $\boldsymbol{\varphi} : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}^3$ is the deformation mapping. The deformation-gradient tensor field $\mathbf{F} : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}^{3 \times 3}$ is defined as $\mathbf{F} := \text{Grad} \boldsymbol{\varphi}(\mathbf{X}, t)$ where $\mathbf{X}$ is the position of a point in the parenchyma at the reference configuration. Let $d\Omega$ be the volume in the current configuration of an infinitesimal volume and $d\Omega_R$ its corresponding infinitesimal volume in the reference configuration. Then, the Jacobian field $J : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}$ is defined as $J := d\Omega/d\Omega_R = \det \mathbf{F}$. Further, we introduce the Green-Lagrange strain tensor $\mathbf{E} : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}^{3 \times 3}$ as

$$\mathbf{E} := \frac{1}{2} (\mathbf{F}^T \mathbf{F} - \mathbf{I}), \quad (1)$$

where $\mathbf{I}$ is the Identity tensor. The Eulerian porosity field $\phi : \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ is defined as the fraction of current air volume $d\Omega^a$ over the overall current infinitesimal volume, that is, $\phi := d\Omega^a/d\Omega$. Then, the air volume at a current point $\mathbf{x}$ is $\phi d\Omega$, and the volume of the alveolar structure is $(1 - \phi) d\Omega$. We also define the Lagrangian porosity field $\Phi : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}$ as $\Phi := d\Omega^a/d\Omega_R$, and we note that both porosity fields are related through the relation $\Phi = J(\phi \circ \boldsymbol{\varphi})$ [48].

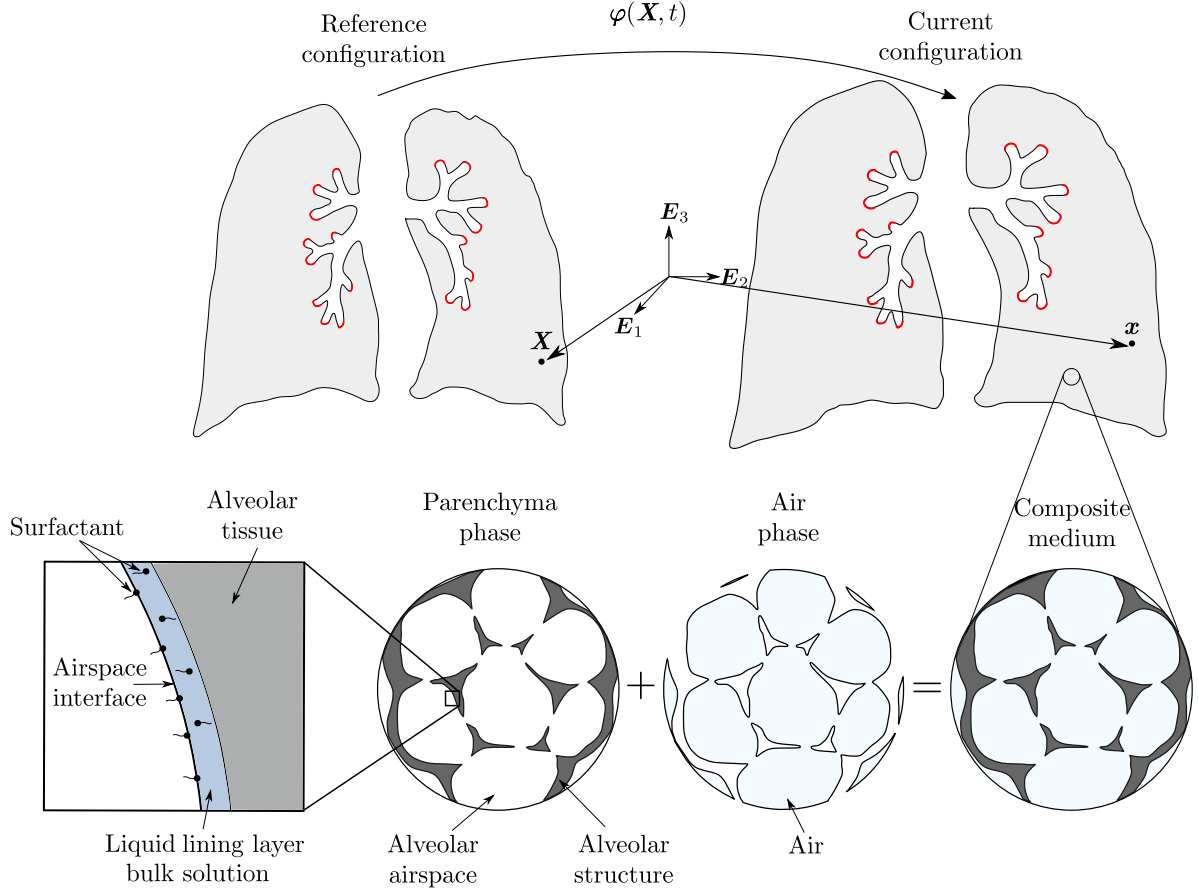


Figure 1: Multiscale approach to model surface tension effects in the lung parenchyma. Lung tissue is idealized as a continuum in which an infinitesimal element of parenchyma and air coexist in a point $\mathbf{x}$ of the current configuration. Lung parenchyma comprises the alveolar structure and alveolar airspace. The alveolar structure is partitioned into the connective tissue and the liquid lining layer that contains surfactant in its bulk solution and on the interface with the air.

Conservation laws in poromechanics are naturally formulated in the current configuration. Let $\rho_a : \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ and $\rho_s : \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ be the air and alveolar structure current density fields, respectively. Further, let $\mathbf{v}_a : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^3$ be the current air velocity fields and $\mathbf{v}_s : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^3$ the current alveolar structure velocity field. The mass conservation for the airspace and alveolar structure are then governed by the expressions [18]

$$\frac{\partial(\rho_a \phi)}{\partial t} + \text{div}(\rho_a \phi \mathbf{v}_a) = 0 \quad \text{in } \Omega \times \mathbb{R}, \quad (2)$$

$$\frac{\partial(\rho_s(1 - \phi))}{\partial t} + \text{div}(\rho_s(1 - \phi) \mathbf{v}_s) = 0 \quad \text{in } \Omega \times \mathbb{R}, \quad (3)$$

respectively. We further define the filtration airflow vector field $\mathbf{q} : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^3$ as

$$\mathbf{q} := \phi(\mathbf{v}_a - \mathbf{v}_s). \quad (4)$$

Then, one can show that air mass conservation expressed relative to the movement of the alveolar structure reads

$$\frac{\partial(\rho_a \phi)}{\partial t} + \text{grad}(\rho_a \phi) \cdot \mathbf{v}_s + \rho_a \phi \text{div} \mathbf{v}_s + \text{div}(\rho_a \mathbf{q}) = 0 \quad \text{in } \Omega \times \mathbb{R}. \quad (5)$$

Let $\boldsymbol{\sigma} : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^{3 \times 3}$ be the Cauchy stress tensor for the composite medium, $\mathbf{b} : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^3$ the current body force density, $\mathbf{a}_a : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^3$ the current acceleration field for the air phase, and $\mathbf{a}_s : \Omega \times \mathbb{R} \rightarrow \mathbb{R}^3$ the current acceleration field alveolar structure phase. Then, the linear momentum balance reads

$$\text{div} \boldsymbol{\sigma} + \rho_s(1 - \phi)(\mathbf{b} - \mathbf{a}_s) + \rho_a \phi(\mathbf{b} - \mathbf{a}_a) = 0 \quad \text{in } \Omega \times \mathbb{R}. \quad (6)$$

We will be interested in expressing all governing balance laws in the reference configuration. To this end, we consider the pull-back of the filtration airflow vector field $\mathbf{Q} : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}^3$, which, by standard arguments, takes the form

$$\mathbf{Q} = J(\mathbf{F}^{-1} \mathbf{q}) \circ \boldsymbol{\varphi}. \quad (7)$$

Then, the pull-back of the air mass balance equation (5) reads

$$\frac{\partial((\rho_a \circ \boldsymbol{\varphi})\Phi)}{\partial t} + \text{Div}((\rho_a \circ \boldsymbol{\varphi})\mathbf{Q}) = 0 \quad \text{in } \Omega_R \times \mathbb{R}. \quad (8)$$

Further, assuming the air phase is incompressible [29], (8) simplifies to

$$\frac{\partial \Phi}{\partial t} + \text{Div} \mathbf{Q} = 0 \quad \text{in } \Omega_R \times \mathbb{R}. \quad (9)$$

The mass balance law for the structural matrix is straightforward because its mass is constant, and reads

$$\frac{\partial((\rho_s \circ \boldsymbol{\varphi})(J - \Phi))}{\partial t} = 0 \quad \text{in } \Omega_R \times \mathbb{R}. \quad (10)$$

We note that in writing (10), we do not restrict mass exchange processes within the alveolar structure, such as those we will consider for pulmonary surfactant later. To obtain a reference form of the linear momentum balance, we define the second Piola-Kirchhoff stress tensor field $\mathbf{S} : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}^{3 \times 3}$ as

$$\mathbf{S} = J \mathbf{F}^{-1} \boldsymbol{\sigma} \mathbf{F}^{-T}. \quad (11)$$

Then, by pulling back (6) and neglecting inertial terms, the reference form of the conservation of linear momentum reads

$$\text{Div}(\mathbf{F} \mathbf{S}) + R \mathbf{B} = 0 \quad \text{in } \Omega_R \times \mathbb{R}, \quad (12)$$

where $\mathbf{B} := \mathbf{b} \circ \boldsymbol{\varphi}$, $R := J(\rho \circ \boldsymbol{\varphi})$ with $\rho := (1 - \phi)\rho_s + \phi\rho_a$.

2.2. Thermodynamics of Tissue Mechanics and Surfactant Dynamics

To ensure thermodynamic consistency in our poromechanical framework, we consider the parenchyma Helmholtz free energy (per unit of reference volume) function Ψ_{par} and we write the Clausius-Duhem inequality for an isothermal porous media as [18]

$$\left(\mathbf{S} : \dot{\mathbf{E}} + P_{\text{alv}} \dot{\Phi} - \dot{\Psi}_{\text{par}} \right) + \left(-\text{Grad}(P_{\text{alv}}) + \rho_a \mathbf{F}^T \mathbf{B} \right) \cdot \mathbf{Q} \geq 0, \quad (13)$$

in which we have assumed that there are no chemical interactions between air and structural alveolar phases, and where $P_{\text{alv}} : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}$ is the air pressure field. We further note that the dot on a field indicates the total time (material) derivative. Equation (13) considers an open system for the air phase and a closed system for the alveolar structure. In addition, the first and second terms in (13) represent the energy dissipation that takes place in the parenchyma and in the air phase due to viscous airflow. Thus, we can separate (13) into two inequalities, one per phase, which read

$$\mathcal{D}_{\text{air}} = \left(-\text{Grad}(P_{\text{alv}}) + \rho_a \mathbf{F}^T \mathbf{B} \right) \cdot \mathbf{Q} \geq 0, \quad (14)$$

$$\mathcal{D}_{\text{par}} = \mathbf{S} : \dot{\mathbf{E}} + P_{\text{alv}} \dot{\Phi} - \dot{\Psi}_{\text{par}} \geq 0. \quad (15)$$

We note that equation (14) imposes a thermodynamic restriction on the constitutive law for airflow. A standard way to ensure its positivity is to consider a quadratic dissipation form, which leads to the classical Darcy's law [13],

$$\mathbf{Q} = J \mathbf{F}^{-1} \mathbf{k} \mathbf{F}^{-T} \left[-\text{Grad}(P_{\text{alv}}) + \rho_a \mathbf{F}^T \mathbf{B} \right], \quad (16)$$

where $\mathbf{k}$ is the permeability tensor, which, for an isotropic flow behavior, reads $\mathbf{k} = k \mathbf{I}$. Similarly, equation (15) restricts the constitutive relation for the parenchyma, and we note that Ψ_{par} admits as natural arguments the state variables $\mathbf{E}$, Φ , in addition to internal variables that govern internal processes. Based on the high water percentage in the alveolar tissue and the liquid lining layer [39], we will assume that the alveolar structure is incompressible. Then, it follows that the Lagrangian porosity field Φ depends on the Jacobian, such that the following relationship holds [18]

$$J - \Phi = 1 - \Phi_R \quad (17)$$

where Φ_R is the reference Lagrangian porosity. Replacing this expression in (15) and using the identity

$$\dot{J} = J \mathbf{F}^{-1} \mathbf{F}^{-T} : \dot{\mathbf{E}}, \quad (18)$$

we rewrite the parenchyma dissipation inequality (15) as

$$\mathcal{D}_{\text{par}} = \mathbf{S} : \dot{\mathbf{E}} + P_{\text{alv}} J \mathbf{F}^{-1} \mathbf{F}^{-T} : \dot{\mathbf{E}} - \dot{\Psi}_{\text{par}} \geq 0. \quad (19)$$

To consider the surfactant-dependent surface tension in our formulation, we will postulate the existence of a parenchyma Helmholtz free energy Ψ_{par} that can split into the classical

hyperelastic energy Ψ^{hyp} associated with the alveolar tissue and the airspace, and into energy Ψ^{st} related to the surface tension that results from surfactant interactions within the liquid lining layer subsystem [76]

$$\Psi_{\text{par}} := \Psi^{\text{hyp}} + \Psi^{\text{st}}. \quad (20)$$

Due to the additive nature of the Helmholtz free energy, we also assume the decomposition [83]

$$\Psi^{\text{st}} := \Psi_{\text{surf}}^{\text{st}} + \Psi_{\text{bulk}}^{\text{st}}, \quad (21)$$

where $\Psi_{\text{surf}}^{\text{st}}$ is the surface energy stored in the airspace interface and $\Psi_{\text{bulk}}^{\text{st}}$ is the energy contribution of the surfactant bulk solution. To incorporate the surfactant exchange (adsorption-desorption) between the bulk solution and the airspace interface, we introduce the interfacial surfactant mass density $M_{\text{surf}} : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}$, defined as $M_{\text{surf}} := d\mathcal{M}_{\text{surf}}/d\Omega_R$, with $d\mathcal{M}_{\text{surf}}$ the mass of (adsorbed) surfactants that covers the airspace interface in an infinitesimal current volume; and $M_{\text{bulk}} : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}$ the bulk solution surfactant mass density, whose definition is analogous to M_{surf} . In addition, to consider effects of area changes over the energy Ψ^{st} , we define the interfacial area density $A : \Omega_R \times \mathbb{R} \rightarrow \mathbb{R}$ as $A := d\mathcal{A}/d\Omega_R$, where $d\mathcal{A}$ is the current area of airspace interface in the deformed infinitesimal volume. We remark that implicit in our internal variable framework, the airspace interface and the bulk solution are described by their mean values, neglecting fluctuations within them. Then, we will assume the validity of the thermodynamic Gibbs relations, and using previous definitions, we can write the energy balance for both components of the liquid lining layer subsystem. For the case of the airspace interface, the energy balance considers the work related to changes in the airspace interfacial area and the energy associated with changes in the interfacial surfactant mass and reads [11, 76]

$$d\Psi_{\text{surf}}^{\text{st}} = \gamma dA + \mu_{\text{surf}} dM_{\text{surf}}, \quad (22)$$

in which γ is the airspace surface tension and μ_{surf} is the (mass-based) chemical potential of the surfactants in the airspace interface. From equation (22), we note that the extensive properties A and M_{surf} are the natural variables for the free energy $\Psi_{\text{surf}}^{\text{st}}$, and we can define the airspace surface tension as the partial derivative of the surface Helmholtz free energy with respect to the interfacial area density [37]

$$\gamma := \frac{\partial \Psi_{\text{surf}}^{\text{st}}}{\partial A}, \quad (23)$$

while its partial derivative concerning the interfacial mass gives the chemical potential

$$\mu_{\text{surf}} := \frac{\partial \Psi_{\text{surf}}^{\text{st}}}{\partial M_{\text{surf}}}. \quad (24)$$

In the case of the incompressible bulk solution, the energy balance is given by

$$d\Psi_{\text{bulk}}^{\text{st}} = \mu_{\text{bulk}} dM_{\text{bulk}}, \quad (25)$$

with μ_{bulk} the chemical potential of surfactants in the bulk solution, which is different from μ_{surf} for a non-equilibrium state, and that can be obtained from the bulk free energy as

$$\mu_{\text{bulk}} := \frac{\partial \Psi_{\text{bulk}}^{\text{st}}}{\partial M_{\text{bulk}}}. \quad (26)$$

Since the surfactant mass exchange is an internal process in the parenchyma, M_{surf} and M_{bulk} can be related through the mass conservation of surfactant in an infinitesimal volume, that reads

$$M_{\text{surf}} + M_{\text{bulk}} = \text{constant} \quad (27)$$

and we note that the total differential of the surfactant bulk mass density in (25) can be rewritten as $dM_{\text{bulk}} = -dM_{\text{surf}}$. Furthermore, noting that for an incompressible alveolar structure phase, variations in A are completely defined by the Green-Lagrange strain tensor $\mathbf{E}$, we conclude that the parenchyma Helmholtz free energy has as arguments $\mathbf{E}$ and M_{surf} , which corresponds to an *internal variable* for the parenchyma energy dissipation. Based on this physicochemical microstructural nature, we define the Helmholtz free energies related to the airspace interface and the surfactant bulk solution as

$$\Psi_{\text{surf}}^{\text{st}} := \Psi_{\text{surf}}^{\text{st}}((A(\mathbf{E}), M_{\text{surf}})), \quad \Psi_{\text{bulk}}^{\text{st}} := \Psi_{\text{bulk}}^{\text{st}}(M_{\text{bulk}}(M_{\text{surf}})) \quad (28)$$

respectively, where $\Psi_{\text{surf}}^{\text{st}}$ depends on the Green-Lagrange strain tensor $\mathbf{E}$ through the interfacial area density A and on the interfacial surfactant mass density M_{surf} ; and $\Psi_{\text{bulk}}^{\text{st}}$ only is a function of M_{surf} through the bulk solution surfactant mass density M_{bulk} . Note that the bulk solution energy $\Psi_{\text{bulk}}^{\text{st}}$ does not depend on $\mathbf{E}$ due to the incompressibility of the alveolar structure. In a more general case, the energy balance (25) has to include a volume-work term, and the bulk solution energy $\Psi_{\text{bulk}}^{\text{st}}$ should also depend on the parenchyma deformations through the current volume of the bulk solution. Following a standard Coleman-Noll procedure [14, 15], we proceed with the calculation of the total derivative of the Helmholtz free energy Ψ_{par} , which, applying the chain rule, result in

$$\dot{\Psi}_{\text{par}} = \frac{\partial \Psi^{\text{hyp}}}{\partial \mathbf{E}} : \dot{\mathbf{E}} + \frac{\partial \Psi_{\text{surf}}^{\text{st}}}{\partial A} \frac{\partial A}{\partial \mathbf{E}} : \dot{\mathbf{E}} + \frac{\partial \Psi_{\text{surf}}^{\text{st}}}{\partial M_{\text{surf}}} \dot{M}_{\text{surf}} + \frac{\partial \Psi_{\text{bulk}}^{\text{st}}}{\partial M_{\text{bulk}}} \frac{\partial M_{\text{bulk}}}{\partial M_{\text{surf}}} \dot{M}_{\text{surf}}. \quad (29)$$

Using definitions (23), (24), (26) and noting from equation (27) that $\partial M_{\text{bulk}} / \partial M_{\text{surf}} = -1$, the total derivative (29) is rewritten as

$$\dot{\Psi}_{\text{par}} = \left(\frac{\partial \Psi^{\text{hyp}}}{\partial \mathbf{E}} + \gamma \frac{\partial A}{\partial \mathbf{E}} \right) : \dot{\mathbf{E}} + \mathcal{L} \dot{M}_{\text{surf}}, \quad (30)$$

where we have defined the thermodynamic force conjugated with the rate $\dot{M}_{\text{surf}}$ as

$$\mathcal{L} := \mu_{\text{surf}} - \mu_{\text{bulk}}. \quad (31)$$

We remark that from a macroscopic perspective, such as the one considered in this work, the terms μ_{surf} and μ_{bulk} should be interpreted as *internal* chemical potentials because they

describe internal interactions within the liquid lining layer subsystem, and not the interaction with other matter such as the air phase [43]. Grouping terms and replacing (30) in (19), the following thermodynamic inequality arises

$$\mathcal{D}_{\text{par}} = \left(\mathbf{S} + P_{\text{alv}} J \mathbf{F}^{-1} \mathbf{F}^{-T} - \gamma \frac{\partial A}{\partial \mathbf{E}} - \frac{\partial \Psi^{\text{hyp}}}{\partial \mathbf{E}} \right) : \dot{\mathbf{E}} - \mathcal{L} \dot{M}_{\text{surf}} \geq 0. \quad (32)$$

We note that equation (32) is similar to the Clausius-Duhem inequality for a poromechanical medium with incompressible solid matrix without interfacial effects [18], plus a term opposite to the pore pressure that takes into account the surface tension; and a dissipative term due to dynamic of the surfactant. In view of classical continuum mechanics arguments [14, 15], the inequality (32) must be satisfied for an arbitrary choices of $\dot{\mathbf{E}}$, and we deduce the constitutive equation

$$\mathbf{S} = \frac{\partial \Psi^{\text{hyp}}}{\partial \mathbf{E}} + \gamma \frac{\partial A}{\partial \mathbf{E}} - P_{\text{alv}} J \mathbf{F}^{-1} \mathbf{F}^{-T}. \quad (33)$$

Furthermore, in our formulation the reference configuration satisfies $\mathbf{S} = 0$, $P_{\text{alv}} = 0$ and $M_{\text{surf}} = M_{\text{surf}}^R$ when $\mathbf{E} = \mathbf{I}$, with M_{surf}^R the reference interfacial surfactant mass. Note that to ensure a stress-free reference configuration, M_{surf}^R should be chosen such that $\gamma = 0$ in the reference configuration. Then, using (33), the dissipative potential due to non-equilibrium internal surfactant interactions remains

$$\mathcal{D}_{\text{par}} = -\mathcal{L} \dot{M}_{\text{surf}} \geq 0. \quad (34)$$

In equation (34), the term $\dot{M}_{\text{surf}}$ must provide an *evolution law* for the interfacial surfactant mass in the form of a differential equation

$$\dot{M}_{\text{surf}} = \mathcal{F}, \quad (35)$$

where $\mathcal{F}$ is a term defined by chosen surfactant material model. We remark that the first and second terms of (33) are related to the main recoil mechanisms of the lung, named the tissue elasticity and the alveolar surface tension, which is one of the essential contributions of this work.

To provide a specific expression of the stress tensor $\mathbf{S}$, we need to compute the geometrical factor $\partial A / \partial \mathbf{E}$ in equation (33). To this end, we idealize the incompressible alveolar structure as a spherical shell whose void porous space (airspace) is filled with air. We remark that the spherical surface defined by the internal radius of the spherical shell represents the airspace interface. Let $\mathcal{R}_{RI} : \Omega_R \rightarrow \mathbb{R}$ and $\mathcal{R}_{RE} : \Omega_R \rightarrow \mathbb{R}$ the inner and outer undeformed radii of the spherical shell. Using basic geometric relationships, the overall reference volume, the reference air volume, and the reference Lagrangian porosity for this microstructure can be expressed by $d\Omega_R = (4/3)\pi \mathcal{R}_{RE}^3$, $d\Omega_R^a = (4/3)\pi \mathcal{R}_{RI}^3$, and

$$\Phi_R = \frac{\mathcal{R}_{RI}^3}{\mathcal{R}_{RE}^3}, \quad (36)$$

respectively. Furthermore, let $r_i : \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ and $r_r : \Omega \times \mathbb{R} \rightarrow \mathbb{R}$ the deformed inner and outer radii, respectively, that are written in the reference configuration as $\mathcal{R}_i = r_i \circ \boldsymbol{\varphi}$ and $\mathcal{R}_e = r_e \circ \boldsymbol{\varphi}$. In a similar way that references volumetric relationships, the overall infinitesimal volume, air volume, and Lagrangian porosity field of the deformed microstructure are written as $d\Omega = (4/3)\pi\mathcal{R}_e^3$, $d\Omega^a = (4/3)\pi\mathcal{R}_i^3$ and

$$\Phi = \frac{\mathcal{R}_i^3}{\mathcal{R}_{RE}^3}, \quad (37)$$

respectively. The interfacial area also can be written as a function of the internal radius of the deformed sphere as $d\mathcal{A} = 4\pi(\mathcal{R}_i)^2$. Then, the interfacial area density is $A = 3\mathcal{R}_i^2/\mathcal{R}_{RE}^3$, which using (36) and (37), can be rewritten as [82]

$$A = \frac{3}{\mathcal{R}_{RI}} \Phi_R^{1/3} \Phi^{2/3}. \quad (38)$$

Through the chain rule, we can compute

$$\frac{\partial A}{\partial \mathbf{E}} = \frac{\partial A}{\partial \Phi} \frac{\partial \Phi}{\partial \mathbf{E}} = \frac{2}{\mathcal{R}_{RI}} \left(\frac{\Phi_R}{\Phi} \right)^{1/3} J \mathbf{F}^{-1} \mathbf{F}^{-T} \quad (39)$$

where we had used the identity

$$\frac{\partial \Phi}{\partial \mathbf{E}} = J \mathbf{F}^{-1} \mathbf{F}^{-T}. \quad (40)$$

Finally, replacing (39) into (33) the total stress tensor is expressed as

$$\mathbf{S} = \frac{\partial \Psi^{\text{hyp}}}{\partial \mathbf{E}} + \frac{2\gamma}{\mathcal{R}_{RI}} \left(\frac{\Phi_R}{\Phi} \right)^{1/3} J \mathbf{F}^{-1} \mathbf{F}^{-T} - P_{\text{alv}} J \mathbf{F}^{-1} \mathbf{F}^{-T}. \quad (41)$$

In equation (41) we define the term

$$P_\gamma = \frac{2\gamma}{\mathcal{R}_{RI}} \left(\frac{\Phi_R}{\Phi} \right)^{1/3} \quad (42)$$

as the collapse pressure due to the surface tension. In this term we note that for a given values of reference external radius $\mathcal{R}_{RE}$ and reference Lagrangian porosity Φ_R , and considering (36) we can write $\mathcal{R}_{RI} := \mathcal{R}_{RI}(\Phi_R, \mathcal{R}_{RE})$ and choose $\mathcal{R}_{RE}$ and Φ_R as geometrical parameters of the model.

2.3. Constitutive models for parenchymal tissue and surfactant

To close our formulation, specific constitutive models for tissue and surfactant are needed. In the case of pulmonary tissue, for simplicity, we adopt an exponential-like material model with an energy density function that reads [21, 45, 74]

$$\Psi^{\text{hyp}} = c \left[\exp(aJ_1^2 + bJ_2) - 1 \right], \quad (43)$$

where a, b, c are material parameters, and J_1, J_2 are the first and second invariant of the Green-Lagrange strain tensor $\mathbf{E}$, i.e.,

$$J_1 := \text{tr} \mathbf{E}, \quad J_2 := \frac{1}{2} [(\text{tr} \mathbf{E})^2 - \text{tr} \mathbf{E}^2]. \quad (44)$$

In the case of pulmonary surfactant, to the best of our knowledge, there are currently no Helmholtz free energy expressions Ψ^{st} suitable for the complex behavior of pulmonary surfactants. We also note that traditional model for surfactants, such as the Langmuir or the Frumkin models, do not consider the phase changes observed in pulmonary surfactants. Furthermore, it is known that in these expressions, surface tension diverges for high interfacial surfactant concentration [49], not allowing surface tensions to decrease their magnitude near zero, a behavior typically observed in pulmonary alveoli. Based on these limitations, we adopt the adsorption-limited model proposed by Otis et al. [58] and integrate it into our continuum poromechanical framework. This surfactant model has three phases whose thresholds are defined regarding the interfacial surfactant concentration Γ , defined as

$$\Gamma := \frac{M_{\text{surf}}}{A}. \quad (45)$$

We introduce the material parameter Γ_{∞} , named the maximum equilibrium interfacial concentration, to define these regimes. For concentration less than Γ_{∞} , surfactant can be adsorbed at the interface or desorbed from it, according to the Langmuir non-equilibrium kinetics. For concentrations between Γ_{∞} and the maximum dynamic concentration Γ_{max} (tightly packing possible), the surfactant behaves as an insoluble compound in which there is no surfactant mass exchange, and any changes in interfacial concentration are only due to changes in the area. When $\Gamma = \Gamma_{\text{max}}$, any area reduction results in the squeeze out of interfacial surfactant to the bulk. For this three-phase model, the term $\mathcal{F}$ in equation (35) is written as

$$\mathcal{F} = \begin{cases} A(k_1 C(\Gamma_{\infty} - \Gamma) - k_2 \Gamma) & \text{if } \Gamma < \Gamma_{\infty}, \\ 0 & \text{if } \{\Gamma_{\infty} \leq \Gamma < \Gamma_{\text{max}}\} \text{ or } \{\Gamma = \Gamma_{\text{max}} \text{ and } \dot{A} \geq 0\}, \\ \Gamma_{\text{max}} \dot{A} & \text{if } \Gamma = \Gamma_{\text{max}} \text{ and } \dot{A} < 0, \end{cases} \quad (46)$$

where k_1, k_2 , and C are material parameters that represent the adsorption coefficient, desorption coefficient, and bulk surfactant concentration, respectively. We note that the deformation-rate dependence of the squeeze-out regime represents a reversible evolution of the internal variable in this regime [44]. To prevent an additional stress term in equation (33), we will neglect the force $\mathcal{L}$ for the stress calculation during this regime. We further write the surface

tension γ as the piecewise function,

$$\gamma = \begin{cases} \gamma_0 - m_1 \left(\frac{\Gamma}{\Gamma_\infty} \right) & \text{if } \Gamma < \Gamma_\infty, \\ \gamma_\infty - m_2 \left(\frac{\Gamma}{\Gamma_\infty} - 1 \right) & \text{if } \{\Gamma_\infty \leq \Gamma < \Gamma_{\max}\} \text{ or } \{\Gamma = \Gamma_{\max} \text{ and } \dot{A} \geq 0\}, \\ \gamma_{\min} & \text{if } \Gamma = \Gamma_{\max} \text{ and } \dot{A} < 0, \end{cases} \quad (47)$$

with m_1 the slope of the Langmuir regime isotherm; m_2 the slope of the insoluble regime isotherm; γ_∞ the surface tension corresponding to Γ_∞ ; and $\gamma_{\min}$ the minimum surface tension that occurs when the dynamic maximum concentration $\Gamma_{\max}$ is reached. Note from (46) that during the Langmuir regime, the interfacial concentration for a stationary process ($\mathcal{F} = 0$) tends to the value

$$\Gamma_{\text{eq}} = \left(\frac{k_1 C}{k_1 C + k_2} \right) \Gamma_\infty. \quad (48)$$

and surface tension during this regime also tends to the equilibrium surface tension defined as $\gamma_{\text{eq}} := \gamma(\Gamma_{\text{eq}})$. Further, we note that for given material parameters $\Gamma_\infty, \gamma_0, \gamma_{\min}, m_1, m_2$, we can write

$$\gamma_\infty = \gamma_0 - m_1, \quad (49)$$

and

$$\Gamma_{\max} = \Gamma_\infty \left(1 + \frac{\gamma_\infty - \gamma_{\min}}{m_2} \right). \quad (50)$$

2.4. Initial boundary value problem and weak formulation

Balance equations and thermodynamic considerations give rise to the governing equations of the poromechanical problem, which in our case reads: Find the deformation mapping field $\boldsymbol{\varphi}$, the alveolar mapping field P_{alv} and the interfacial surfactant mass density field M_{surf} such that

$$\text{Div}((\mathbf{F}\mathbf{S}) + R\mathbf{B}) = \mathbf{0} \quad \text{in } \Omega_R \times \mathbb{R}, \quad (51)$$

$$\frac{\partial \Phi}{\partial t} + \text{Div}(\mathbf{Q}) = 0 \quad \text{in } \Omega_R \times \mathbb{R}, \quad (52)$$

$$\frac{\partial M_{\text{surf}}}{\partial t} - \mathcal{F} = 0 \quad \text{in } \Omega_R \times \mathbb{R}, \quad (53)$$

where the second Piola-Kirchhoff stress $\mathbf{S} = \mathbf{S}(\text{Grad } \boldsymbol{\varphi}, P_{\text{alv}}, M_{\text{surf}})$ is given by (41), the airflow follows the Darcy law (16), and the evolution law for the interfacial surfactant mass density field is given by (46). Equations (51) and (52) are complemented by boundary

conditions

$$\boldsymbol{\varphi} = \bar{\boldsymbol{\varphi}} \quad \text{on } \partial\Omega_{R_\varphi} \times \mathbb{R}, \quad (54)$$

$$(\mathbf{FS}) \cdot \mathbf{N} = \bar{\mathbf{T}} \quad \text{on } \partial\Omega_{R_T} \times \mathbb{R}, \quad (55)$$

$$P_{\text{alv}} = \bar{P} \quad \text{on } \partial\Omega_{R_P} \times \mathbb{R}, \quad (56)$$

$$\mathbf{Q} \cdot \mathbf{N} = \bar{\mathbf{Q}} \quad \text{on } \partial\Omega_{R_Q} \times \mathbb{R}. \quad (57)$$

with $\mathbf{N}$ is the unitary vector normal to the reference configuration boundary $\partial\Omega_R$ and where we had assumed the partition [41, 71, 79]

$$\partial\Omega_{R_\varphi} \cup \partial\Omega_{R_T} = \partial\Omega_R \quad \partial\Omega_{R_\varphi} \cap \partial\Omega_{R_T} = \emptyset, \quad (58)$$

$$\partial\Omega_{R_P} \cup \partial\Omega_{R_Q} = \partial\Omega_R \quad \partial\Omega_{R_P} \cap \partial\Omega_{R_Q} = \emptyset, \quad (59)$$

with $\partial\Omega_{R_\varphi}$, $\partial\Omega_{R_T}$, $\partial\Omega_{R_P}$, and $\partial\Omega_{R_Q}$ the boundaries of prescribed deformation mapping, traction, alveolar pressure, and alveolar airflow, respectively. The formulation of our transient poromechanical problem is completed with the initial conditions for the primary variables $\boldsymbol{\varphi}$, P_{alv} and for the internal variable M_{surf}

$$\boldsymbol{\varphi} = \boldsymbol{\varphi}^0 \quad \text{in } \Omega_R, \quad t = 0, \quad (60)$$

$$P_{\text{alv}} = P_{\text{alv}}^0 \quad \text{in } \Omega_R, \quad t = 0, \quad (61)$$

$$M_{\text{surf}} = M_{\text{surf}}^0 \quad \text{in } \Omega_R, \quad t = 0. \quad (62)$$

Following a standard finite-element procedure, we introduce the solution function spaces

$$\mathcal{S}_\varphi := \{\boldsymbol{\varphi} \mid \boldsymbol{\varphi} \in H^1(\Omega_R, \mathbb{R}^3); \boldsymbol{\varphi} = \bar{\boldsymbol{\varphi}} \text{ on } \partial\Omega_{R_\varphi}\}, \quad (63)$$

$$\mathcal{S}_P := \{P_{\text{alv}} \mid P_{\text{alv}} \in H^1(\Omega_R, \mathbb{R}); P_{\text{alv}} = \bar{P} \text{ on } \partial\Omega_{R_P}\}, \quad (64)$$

$$\mathcal{S}_M := \{M_{\text{surf}} \mid M_{\text{surf}} \in L^2(\Omega_R, \mathbb{R})\} \quad (65)$$

and the associated test function spaces

$$\mathcal{V}_\varphi := \{\delta\boldsymbol{\varphi} \mid \delta\boldsymbol{\varphi} \in H^1(\Omega_R, \mathbb{R}^3); \delta\boldsymbol{\varphi} = \mathbf{0} \text{ on } \partial\Omega_{R_\varphi}\}, \quad (66)$$

$$\mathcal{V}_P := \{\delta P_{\text{alv}} \mid \delta P_{\text{alv}} \in H^1(\Omega_R, \mathbb{R}); \delta P_{\text{alv}} = 0 \text{ on } \partial\Omega_{R_P}\}, \quad (67)$$

$$\mathcal{V}_M := \{\delta M_{\text{surf}} \mid \delta M_{\text{surf}} \in L^2(\Omega_R, \mathbb{R})\}. \quad (68)$$

Multiplying (51), (52) and (53) by their respective test functions, integrating into Ω_R , applying the integration by parts theorem when it is appropriate and using the boundary conditions (55), (57), we obtain the following weak statement

$$\int_{\Omega_R} \mathbf{FS} : \text{Grad } \delta\boldsymbol{\varphi} \, dV - \int_{\Omega_R} R\mathbf{B} \cdot \delta\boldsymbol{\varphi} \, dV - \int_{\partial\Omega_{R_T}} \bar{\mathbf{T}} \cdot \delta\boldsymbol{\varphi} \, dS = 0 \quad (69)$$

$$\int_{\Omega_R} \frac{\partial\Phi}{\partial t} \delta P_{\text{alv}} \, dV - \int_{\Omega_R} \mathbf{Q} \cdot \text{Grad } \delta P_{\text{alv}} \, dV + \int_{\partial\Omega_{R_Q}} \bar{\mathbf{Q}} \delta P_{\text{alv}} \, dS = 0 \quad . \quad (70)$$

$$\int_{\Omega_R} \left\{ \frac{\partial M_{\text{surf}}}{\partial t} - \mathcal{F} \right\} \cdot \delta M_{\text{surf}} \, dV = 0 \quad (71)$$

for all $(\delta\boldsymbol{\varphi}, \delta P_{\text{alv}}, \delta M_{\text{surf}}) \in \mathcal{V}_{\varphi} \times \mathcal{V}_P \times \mathcal{V}_M$. Equation (71) was solved pointwise since gradients of M_{surf} are not involved [30, 32]. Temporal discretization of the weak formulation was carried out using a backward-Euler time integration scheme. We implement the temporally discretized problem in the finite element FEniCS library [42].

3. Numerical simulations

3.1. Model setup and post-processing

The lung domain was segmented from a computed tomography image taken from a healthy volunteer under resting conditions [31], see Figure 2(a) for an axial slice. The computational tetrahedral mesh was generated using the CGAL library [75]. We considered this mesh to be the reference stress-free configuration Ω_R , see Figure 2(b). The lung surface was partitioned into the airway boundaries ($\partial\Omega_{R_{\text{aw}}}$) and the visceral pleura surface ($\partial\Omega_{R_{\text{pl}}}$). The visceral pleura surface was assumed isolated, i.e., no-flux boundary conditions for the flow field. The airway boundary allowed airflow, either by prescribing pressures or flow [4]. Considering that the focus of this work was to model and validate the effect of surface tension on lung mechanics, our simulations considered an excised lung, and therefore the lung surface was free to displace without any mechanical restraint. This condition is consistent with the *in vitro* experimental setup of the supersyringe method [52], which we explain below. Accordingly, the visceral pleura surface was assumed to have zero-traction boundary conditions. Null displacements were prescribed over the airway boundary to prevent rigid-body modes. Throughout our experiments, we assumed that the reference and initial configuration match, which implies $\boldsymbol{\varphi}^0 = \mathbf{0}$, $P_{\text{alv}}^0 = 0$ and $M_{\text{surf}}^0 = M_{\text{surf}}^R$. In particular, we set $M_{\text{surf}}^R = \Gamma_{\text{max}} A^0(\mathbf{X})$, with A^0 the initial alveolar area density field computed directly from the reference alveolar geometry, and Γ_{max} the value of the interfacial surfactant concentration such that the minimum surface tension was zero. The values for Γ_{max} and other parameters assumed for a normal lung, also called an *air-filled lung*, are shown in Table 1 along with references where these values were taken from. In addition to modeling a normal lung, we consider the case in which the effects of surface tension were deactivated ($\gamma = 0$), which is experimentally materialized by supersyringe tests on *saline-filled lungs* [5, 56].

A standard approach to studying lung mechanics is the construction of quasi-static pressure-volume (P-V) curves from the supersyringe experiment [25]. In this work, we used the proposed lung model to construct P-V curves by simulating a supersyringe experiment. During the inflation stage, a total air volume V_T was supplied step by step in N_s regular steps. Each step was divided into two phases: a flow phase, with duration T_f and a pause phase of duration T_p . During the flow phase, air was supplied by imposing a constant flow $\bar{Q} = V_s/(A_{\text{aw}}T_f)$ where $V_s = V_T/N_s$ is the volume supply in each step and A_{aw} is the area of the airways boundary computed as the surface integral over this boundary. Then, flow was set to zero throughout the pause phase. After reaching the target volume, the deflation process was carried out using an analogous procedure, expelling the air step by step. During the entire experiment, we computed the temporal evolution of the volume supplied by

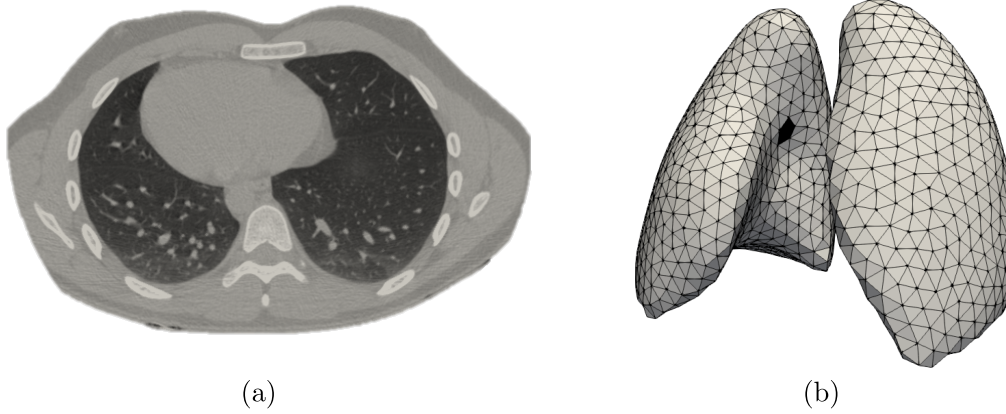


Figure 2: Construction of an anatomical finite-element model of the lungs: (a) Computed-tomography image of a normal human lung at end of expiration, assumed as the reference configuration. (b) Tetrahedral mesh considered in finite element simulations. The airway boundary for the right lung is shown in black. The left airway boundary is omitted for the clarity of the figure.

Parameter	Value	Units	Reference
Γ_∞	$3.0 \cdot 10^{-9}$	g/mm^2	[54]
$\gamma_{\min}$	0	N/mm	[60]
γ_0	$70 \cdot 10^{-6}$	N/mm	[58]
m_1	$47.8 \cdot 10^{-6}$	N/mm	[58]
m_2	$140 \cdot 10^{-6}$	N/mm	[54, 58]
k_1	$1.6 \cdot 10^8$	$\text{mm}^3/(\text{g} \cdot \text{s})$	[19, 58]
k_2	0.016	$1/\text{s}$	[19, 58]
C	$7.3 \cdot 10^{-9}$	g/mm^3	[19, 58]
R_{RE}	0.11	mm	[35]*
Φ_R	0.74	-	[35]*
a	0.433	-	[74]
b	-0.61	-	[74]
c	2.5	kPa	[74]
k	10^4	$\text{mm}^2/(\text{kPa} \cdot \text{s})$	[4, 9]

Table 1: Model parameters considered in numerical simulations. Note that for saline-filled lung simulations, only a, b, c and k are necessary. *Values were computed from the alveolar dimensions given in the reference.

the syringe, also called lung volume (V), the average airway pressure (P), and the average surface tension (γ), defined as

$$V(t) := \int_{\Omega_R} (J - 1) dV, \quad P(t) := \frac{1}{A_{\text{aw}}} \int_{\partial\Omega_{R_{\text{aw}}}} P_{\text{alv}} dS, \quad \gamma(t) := \frac{1}{V_R} \int_{\Omega_R} \gamma dV \quad (72)$$

where V_R is the reference lung volume obtained as $V_R = \int_{\Omega_R} dV$. From these data, the quasi-static pressure-volume and surface tension-volume (γ - V) curves were constructed connecting

the last point of each step. We set $T_f = 0.3$ s , $T_p = 2.0$ s and $V_s = 0.1$ L in our supersyringe simulations. Three different volume excursions for the air-filled lung were simulated: a baseline volume of $V_T = 1.5$ L, a low volume of $V_T = 0.5$ L and a high volume of $V_T = 2.5$ L. Quasi-static curves for the saline-filled lung were also constructed for the baseline volume.

Volume-controlled mechanical ventilation (VCV) simulations were also performed for the air-filled and saline-filled lungs; we refer the reader to Avilés-Rojas and Hurtado [4] for further details of the simulation protocol. During the first phase of this ventilation mode, i.e., inspiration phase, a tidal volume V_{tidal} was supplied by prescribing a constant airflow $\bar{Q} = V_{\text{tidal}}/(A_{\text{aw}}T_{\text{insp}})$ on the airway boundary, with T_{insp} the duration of the inspiratory phase. After this, a pause phase was simulated, during which a null flow was prescribed on the airway boundaries. Next, the expiration was carried out by setting the airway pressure equal to zero. In addition to the lung volume, airway pressure, and surface tension given by (72), the flow of air was computed as $\dot{V}(t) = \partial V(t)/\partial t$. For our VCV experiments, we considered an inspiratory phase of 1.5 s in which a tidal volume of 0.5 L was supplied, followed by 0.75 s of pause and an expiratory phase of 4.5 s.

3.2. Results

Figure 3 shows the temporal evolution of the lung volume, airway pressure, and average surface tension for the normal (air-filled lung) and the deactivated surface-tension (saline-filled lung) cases. The evolution of pressure and surface tension for the normal lung strongly differed between the inflation and deflation branches. For example, during inflation, the surfactant behavior changed from an insoluble regime to a regime governed by Langmuir kinetics at $t = 9.8$ s, in which the surface tension tended to equilibrium, remaining in that regime until the end of the inflation phase. In contrast, during the deflation phase, this regime was quickly abandoned, passing to an insoluble state and finally producing squeeze-out phenomena when the minimum surface tension was reached during the dynamic compression of the syringe phase.

The quasi-static pressure-volume curves constructed from the volume and pressure time series are reported in Figure 4(a). The inflation and deflation limbs for the saline-filled lung case coincide and are nearly linear in the range of volume and pressure considered. In large contrast, in the air-filled lung simulation, the inflation and deflation limbs considerably differ, resulting in a markedly hysteretic behavior. Changes in the slope of P-V curves, also known as lung compliance, were observed during inflation and deflation. Following the notation given in Takeuchi et al. [73], the different slopes in the P-V curve are denoted as C_{start} (starting compliance), C_{inf} (inflation compliance), C_{top} (top compliance), C_{def} (deflation compliance), and the compliance transitioned points as LIP (lower inflection point) and PMC (point of maximum curvature). Figure 4(b) reports the surface tension-volume curve. For the air-filled lung, the γ -V curve displayed a non-linear behavior.

To study the spatial distribution of volumetric changes and the gas fraction, we report the Jacobian and the Eulerian porosity fields in the air-filled and saline-filled lung simulations, for

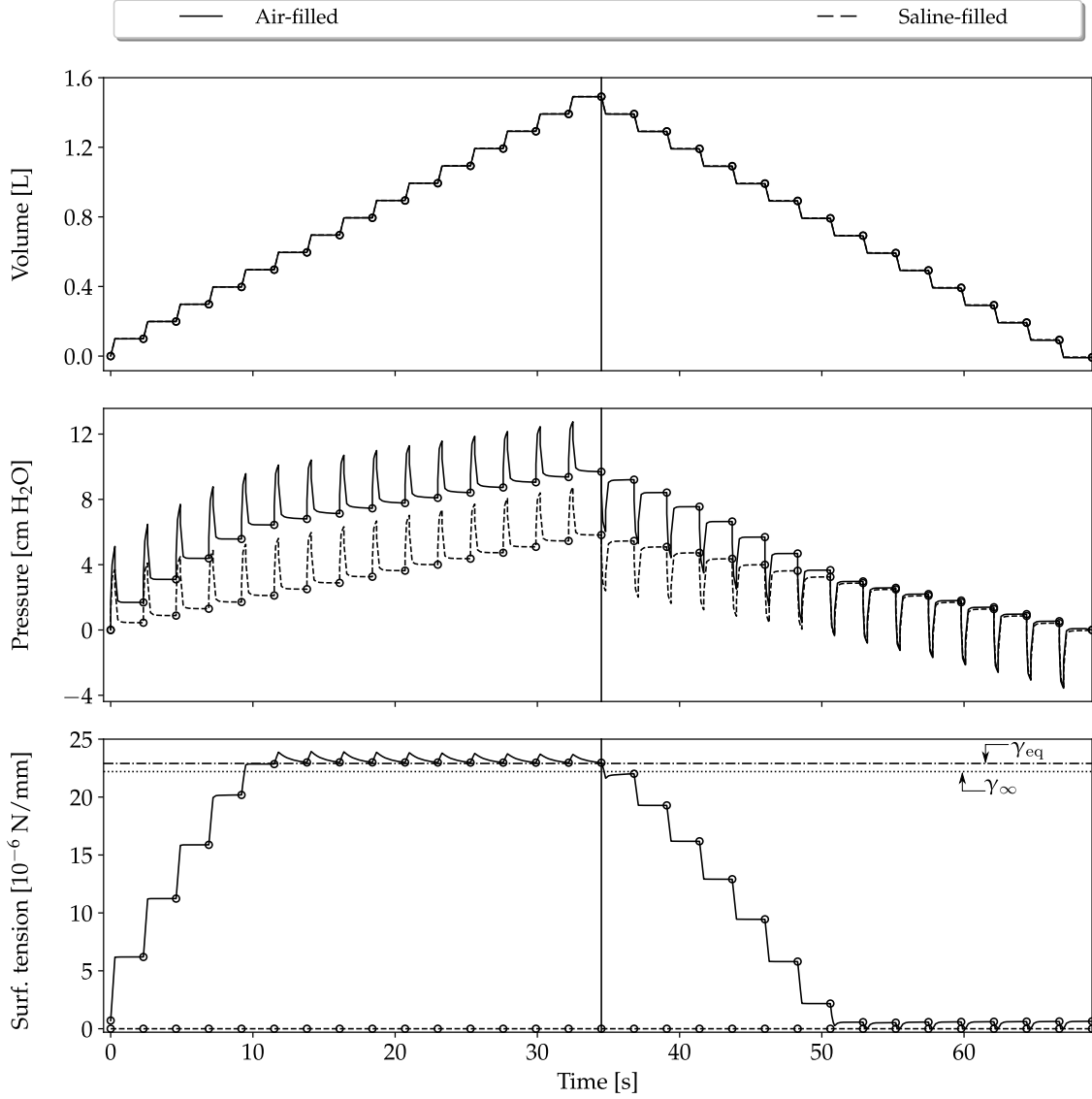


Figure 3: Simulation of supersyringe method for air-filled and saline-filled lungs. Temporal evolution of lung volume, airway pressure, and surface tension is shown for both cases. Black dots indicate the steady-state time instants considered in the construction of quasi-static pressure-volume curves.

airway pressures of 2 and 5 cm H₂O during inflation and deflation, see Figures 5(a) and (b). Figure 5 also reports the tissue hydrostatic stress field, defined as $\sigma_{\text{tissue}}^{\text{hyd}} = (1/3)\text{tr}(\sigma_{\text{tissue}})$ with $\sigma_{\text{tissue}} = J^{-1}\mathbf{F}(\partial\Psi^{\text{hyp}}/\partial\mathbf{E})\mathbf{F}^T$. The Jacobian, Eulerian porosity, and hydrostatic stress fields for the air-filled lung resulted in lower values during inflation than deflation. For example, for an airway pressure of 5 cm H₂O, Eulerian porosity reached a maximum of 0.77 during inflation, while this value increased to 0.81 during deflation. In contrast, no differences were observed during inflation and deflation in the saline-filled lung simulation.

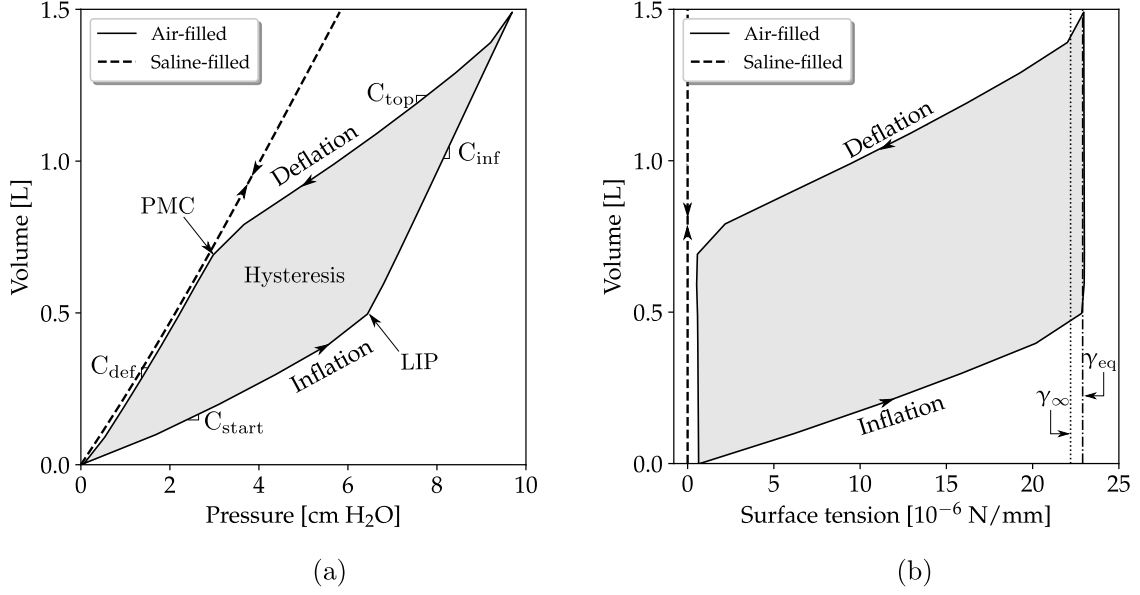


Figure 4: Quasi-static pressure-volume curves for air-filled and saline-filled lungs: (a) pressure-volume curves and (b) surface tension-volume curves. Slopes in the P-V inflation limb are indicated as C_{start} and C_{inf} , while during deflation limb as C_{top} and C_{def} . The lower inflection point (LIP) and point of maximum curvature (PMC) correspond to breakpoints in the P-V curve where slope abruptly changes during inflation and deflation, respectively. The shaded area in the P-V curve indicates the energy dissipation or quasi-static hysteresis due to the surface tension effect of pulmonary surfactant.

Figure 6(a) shows the air-filled pressure-volume curves for the low, baseline, and high volume cases. The high-volume case displayed larger hysteresis compared to other cases, represented by the area enclosed by inflation and deflation limbs. Interestingly, the change in compliance from C_{start} to C_{inf} is similar for the baseline and high volume simulations. In contrast, changes in hysteresis and slope in the low-volume case were negligible. Figure 6 (b) shows the surface tension-volume curve for the three volume cases. While the surface tension for the low-volume simulation seems to remain under an insoluble regime ($\gamma < \gamma_{\infty}$), the other cases exceed the value of γ_{∞} , moving to a Langmuir regime, in which the surface tension tends to equilibrium γ_{eq} .

Figure 7(a) shows the airway pressure, flow, volume, and surface tension signals calculated for the VCV simulations. In this experiment, volume and flow signals during the inspiratory phase are the same for air-filled and saline-filled lung simulations. Maximum inspiratory pressure was higher in the air-filled lung than in the saline-filled lung. We also observed marked differences in flow and volume waveforms during expiration phase. Figures 7(b) and 7(c) show the corresponding dynamic pressure-volume curves and the dynamic surface tension-volume curves, respectively. Both dynamic pressure-volume curves exhibit hysteresis, even though the dynamic surface tension-volume curves do not.

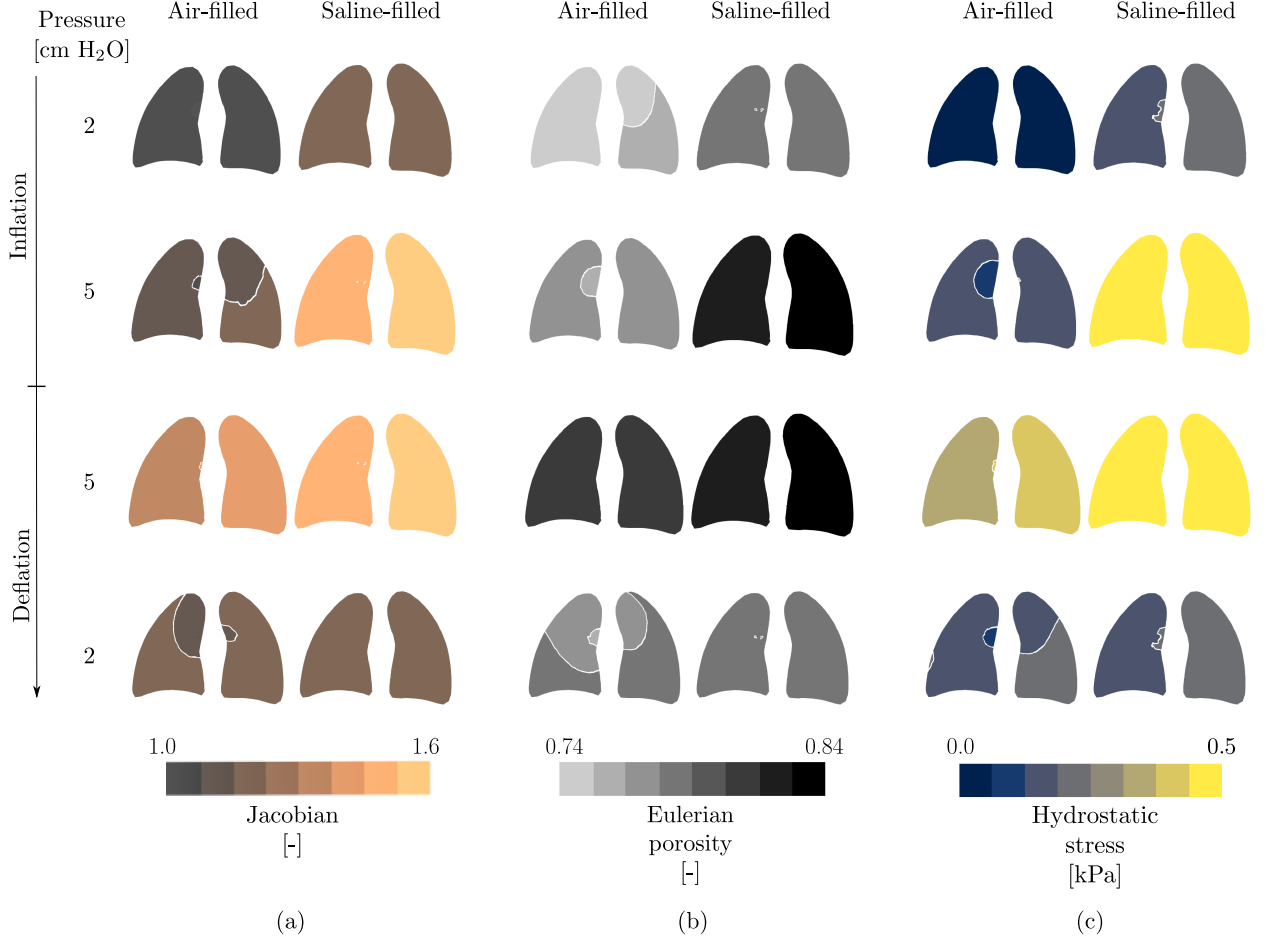


Figure 5: Three-dimensional fields at different pressure levels during the supersyringe method simulation: (a) Jacobian volume-change field, (b) Eulerian porosity field, (c) tissue hydrostatic stress field.

4. Discussion

In this work, we introduce a continuum poromechanical model of lung mechanics that seamlessly incorporates the effect of surfactant dynamics and the resulting alveolar surface tension on the tissue and organ response. Previous continuum formulations of lung mechanics only considered the hyperelastic tissue response and its interaction with gas viscous flow, resulting in standard poroelastic models [4, 9, 29, 59]. To the best of our knowledge, our work constitutes the first attempt to directly address interfacial phenomena resulting from the thin liquid layer surrounding the alveolar tissue and in contact with the air in a continuum formulation, whose impact on pulmonary function is crucial. Due to the scale differences between surfactant and whole organ, surfactant activity was naturally considered in a continuous framework using an internal variables theory approach. This approach allows us to track the surfactant mass in the interface and the consequent transient behavior of surface tension during multiple-step experiments as the supersyringe method; see

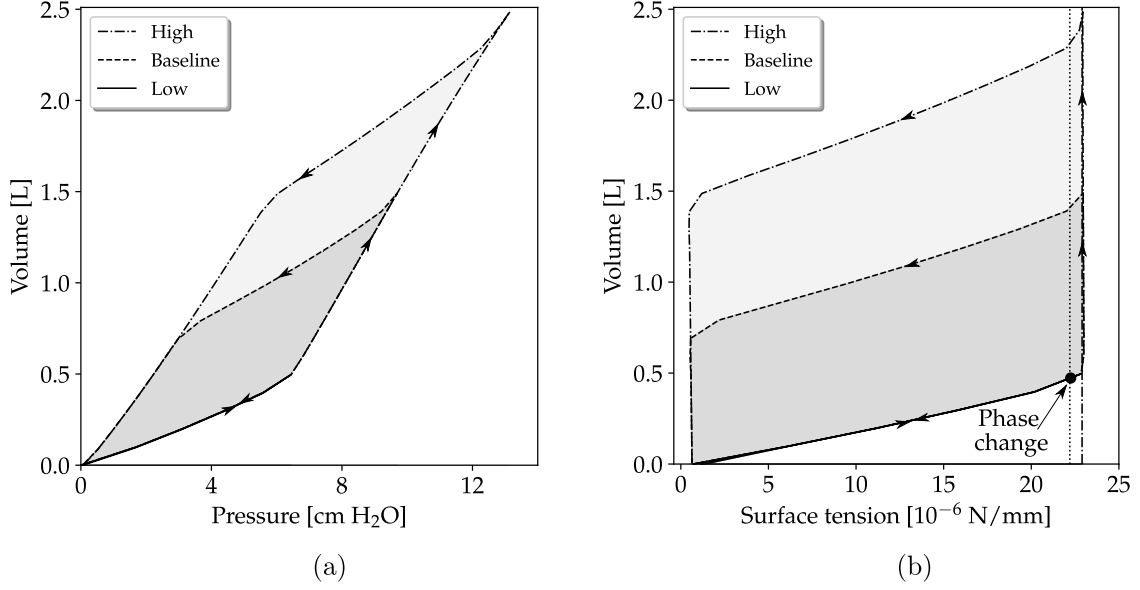


Figure 6: Influence of total excursion volume on quasi-static lung response: (a) pressure-volume curves and (b) surface tension-volume curves. No hysteresis is observed for the low-volume case. A black dot on the $\gamma - V$ curve indicates when, on average, the behavior of the pulmonary surfactant changes from an insoluble regime to one governed by Langmuir kinetics.

the surface tension evolution in Figure 3. Further, via a standard Coleman-Noll procedure and taking the mass of the surfactant as a new state variable, we obtain an expression of Clausius-Duhem inequality similar to the usual one for poroelastic media with incompressible solid matrix [18], plus two new terms: a term opposite to pore pressure that takes into account surface tension; and a dissipative term due to the adsorption-desorption of the surfactant, see equation (32). This expression results in a stress tensor formulation that integrates tissue elasticity and surface tension, the main mechanisms of lung recoil [5, 81]. Remarkably, the surface tension-related term is essentially similar to the classical Young-Laplace equation but modulated by a geometric factor representing the porous nature of alveoli, see (41). Based on this result, we anticipate that the zero-thickness bubble alveolar model traditionally used [69] could be suitable for highly porous alveolar tissue (high value of Φ_R). However, it may not be suitable to take into account the effect of surface tension in lungs with low porosity (low gas fraction), as collapsing pressure P_γ increases when the Lagrangian porosity decreases [57].

The pressure-volume curve is a gold-standard tool to characterize the mechanical properties of the lung, providing critical information to manage the treatment of patients with respiratory distress during mechanical ventilation treatment [1, 25]. Remarkably, our computational model allows the construction of these curves, recovering many features of representative P-V curves observed in medical research and clinical practice [62, 73]. Notably, our results suggest that the P-V inflation and deflation limbs are strongly influenced by the

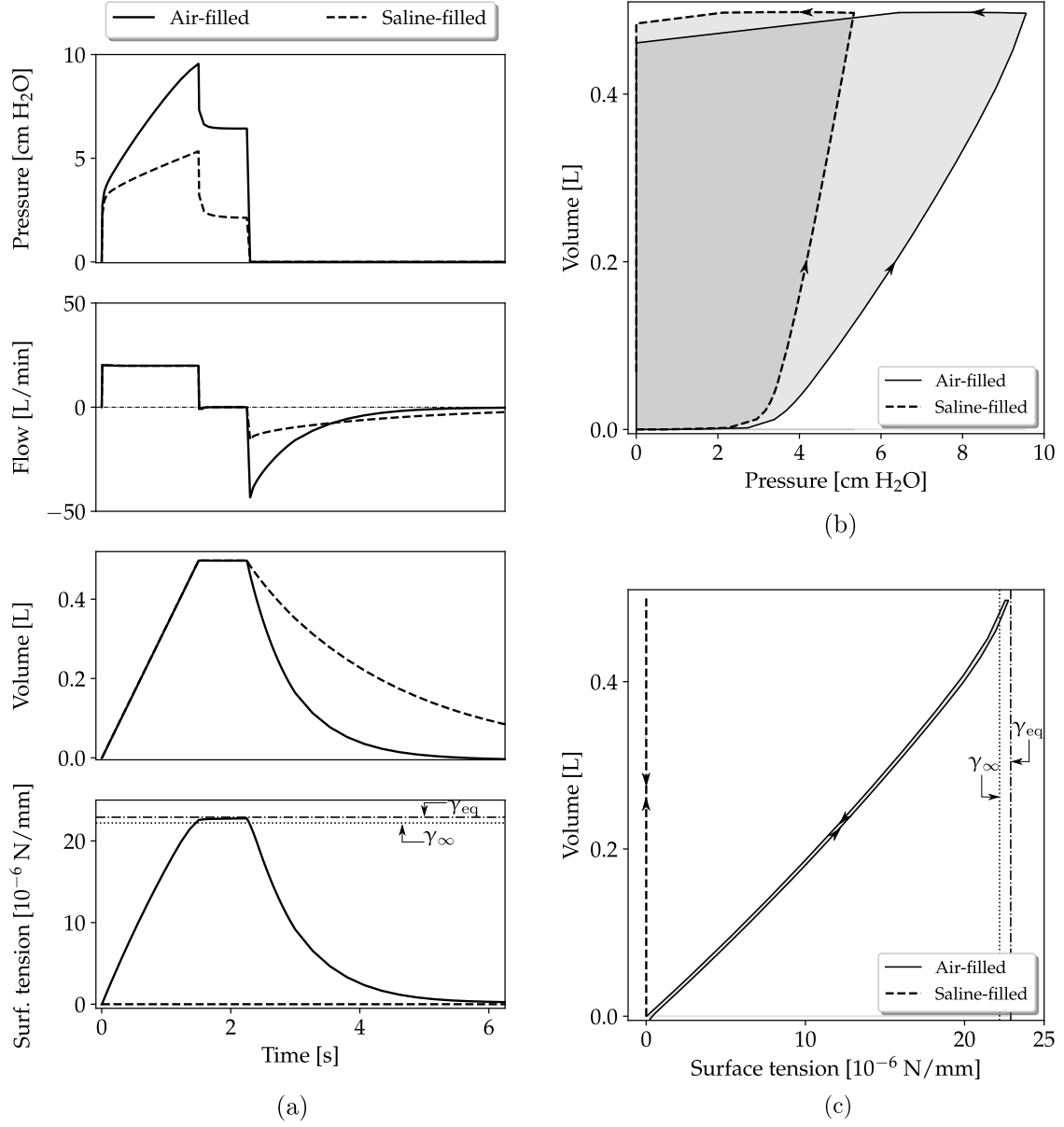


Figure 7: Volume-controlled mechanical ventilation simulations of air-filled and saline-filled lungs: (a) airway pressure, flow, lung volume, and surface tension signals, (b) dynamic pressure-volume curve, and (c) dynamic surface tension-volume curve.

shape of the surface tension-volume curve ($\gamma - V$), see in Figure 4(b). For example, during the loading phase (inflation), surface tension increases with volume, in a similar way to the pressure, until the equilibrium surface tension (γ_{eq}) is reached. In particular, we note that the volume when the surface tension reaches the equilibrium coincides with the improvement in compliance during inflation. We remark that in clinical practice, this transition from low to high compliance is known as the lower inflection point (LIP) and is used as a guide to

set the optimal value of the positive end-expiratory pressure (PEEP) in mechanically ventilated patients because it has been historically related to the recruitment of alveolar units [25, 36, 72]. Interestingly, our predicted LIP is mainly attributed to the dynamics of the surfactant, which transitions from an initially compressed state of very low compressibility (change in surface area per change in surface tension) to a regime with high surface compressibility [68]. We remark that our results are consistent with the experimental results of Smith and Stamenovic [68] (see, for example, Figure 6 of that reference), which show that the shape of P-V curves is directly determined by the molecular properties of the surfactant film relating to the non-equilibrium behavior and adsorption-desorption regime. Our results capture the critical role of surface tension in establishing trend changes and breakpoints in P-V curves, as supported by previous contributions in the literature [28], which along with alveolar recruitment, constitutes one of the main drivers of lung compliance [36].

One of the most remarkable features of our model is its ability to capture dissipative phenomena due to interfacial dynamics such as the quasi-static hysteresis, a common feature observed in quasi-static pressure-volume curves of healthy and diseased lung [22, 50], see the (air-filled) P-V hysteretic loop shown in Figure 4(a). In effect, for the same change in volume, higher pressures are necessary during inflation than deflation, which implies that a greater amount of mechanical work must be done to open the lung than to close it [26]. This quasi-static hysteresis can be attributed to the asymmetry of lung surfactant film behavior during inflation and deflation [23, 70], as Figure 3 shows. Consequently, different pressures are needed for inflation and deflation, and the energy applied during inflation is not fully recovered in the deflation process [20]. Our spatial distribution field predictions also capture the hysteresis-related differences between inflation and deflation stages for the air-filled lung; see Figure 5. In particular, for the same pressure level, the Jacobian distributions suggest that alveolar units are more open during deflation than inflation, which implies a higher gas fraction in the deflation stage, as the Eulerian porosity field shows. As we expected, in saline-filled lungs there are no differences between inflation and deflation limbs for the same pressure level. Furthermore, we note that the absence of collapsing pressure due to surface tension results in larger volumetric changes and tissue hydrostatic stresses than in the air-filled lung case. We remark that quasi-static hysteresis of the whole lung has been previously predicted by discrete models, as proposed by Roth et al. [65]. However, in that model, alveolar ducts were modeled using a four-element Maxwell model [34] with viscoelastic parameters calibrated from previous numerical P-V curves [19], not allowing a direct relationship of the lung mechanics with surfactant properties. Recently, a whole lung acinar units computational model that incorporated the dynamic surfactant physicochemical behavior in the air-liquid interface and the resulting (Laplace) collapsing pressure was proposed [46, 47]. The authors report dynamic curves for low tidal volume breathing but not the quasi-static pressure-volume curves that are usually used to quantify the surfactant contribution to hysteresis. In contrast, here we show that our poromechanical framework can represent the quasi-static hysteretic lung behavior, which is directly influenced by the dynamics of surfactant at the alveolar level.

The influence of volume excursion in the shape of pressure-volume curves was also analyzed; see Figure 6. We note that, while for long excursions, it is possible to identify the lower inflection point on the inflation limb and the point of maximum curvature on the deflation limb, which indicate the changes in compliance during inflation and deflation, respectively, no marked compliance changes are observed for small volumes. Further, we note that the LIP value and the start compliance C_{start} for the medium and longest volumes are similar. These observations suggest that, while the LIP depends on the total volume and does not exist for low volumes, C_{start} and the LIP value (if any) are not influenced by the total volume. Interestingly, the PMC is strongly affected by volume, having a high-pressure value for longer excursions. Notably, many of our volume-dependent predictions are in line with the observations from experimental measurements performed by Takeuchi et al. [73]. Another notable result of our model is the effect of the volume excursion on the difference between the inspiratory and expiratory limbs, which results in larger hysteresis for larger volumes, being negligible for small volumes. We note that this fact has been previously reported for quasi-static P-V curves for human lungs, which suggest that under small deformations, the lung behavior is similar to an ideal elastic (reversible) medium and that non-reversibility phenomena are produced during deep volume cycles, resulting in the lung hysterical behavior [52]. Remarkably, our thermodynamic framework suggests that the value of γ_{∞} , that is, the surface tension that triggers the surfactant phase-change towards the Langmuir regime, corresponds to a threshold that separates a purely elastic domain and a dissipative one. In this sense, it plays a similar role to the yield stress observed in plasticity theory. In effect, surfactant-related dissipated energy would only exist if the non-equilibrium (Langmuir) regime is achieved, leaving an almost elastic behavior for volume levels that do not experience the phase change from insoluble to Langmuir regime.

The applicability of the proposed framework to clinical situations was studied by means of simulating mechanical ventilation protocols typically used in patients with lung failure. Remarkably, our lung simulations were able to predict the evolution of key respiratory signals such as airway pressure, lung flow, and lung volume, see figure 7(a), recovering waveforms readily observed in the clinical practice. For example, the airway pressure increases throughout inspiration due to the square wave of prescribed flow and is then reduced during the pause phase until reaching a plateau pressure, which tends to a steady state. In addition, the negative flow peak and its exponential decay during expiration, usually observed in patients under mechanical ventilation [6], also is recovered. Remarkably, our model shows that the surface tension seems to have an essential influence on the magnitude and shape of these physiological signals. In particular, for the same change in lung volume, the peak inspiratory and plateau pressure are higher for the air-filled case. The higher elastic recoil pressure in air-filled lungs can explain these larger values. Indeed, the elastic recoil of the saline-filled lung is completely explained by the lung tissue elasticity and is only a fraction of the recoil of the normal lung that also considers the collapsing pressure due to surface tension P_{γ} [38]. Consequently, a higher magnitude of the maximum expiratory flow and a lower exponential decay time constant are predicted when surface tension is included. Interestingly, even though the surface tension is neglected in the saline-filled situation, a hysteretical loop in the

dynamic pressure-volume curve is produced. We attribute this dissipative behavior to the airflow in the lung parenchyma; see equation 14. A further qualitative comparison between the areas enclosed by the dynamic pressure-volume curves in Figure 7(b) shows that most of the dynamic hysteresis arises due to flow resistance rather than surface tension. In fact, the contribution of surfactant dynamics to lung hysteresis during transient breathing cycles with low tidal volumes is negligible [67, 70].

This work corresponds to one of the first attempts to incorporate surface tension into a continuous lung model, offering many opportunities for improvement. First, our simulations assumed an unstressed reference configuration based on the geometry of the lung at the end of expiration. This assumption is limited in that prestresses are present during resting conditions due to effect of pleural pressure acting over the lung surface. Some works in the literature have addressed this limitation by estimating the unloaded configuration using fixed-point techniques [9, 59]. However, the experimental validation of methods for the determination of prestress fields remains an open problem. Second, despite the fact that we have captured the inelastic effects linked to surfactant-dependent surface tension, the viscoelastic response of alveolar tissue has not been considered. While hysteresis of surfactant is negligible for normal tidal breathing, tissue viscoelasticity could result in hysteretical loops during these low-volume regimes [67, 70], which is not reflected in our simulations. We note that tissue viscoelasticity could be incorporated in our continuum model using finite-deformation viscoelastic material models, which adds to a non-equilibrium stress term to the expression of the second Piola Kirchhoff tensor [10, 27]. Third, conducting and respiratory airways that compose the complex airway tree structure inside the lung have not been explicitly modeled. Instead, the right and left primary bronchial tubes have been considered a boundary of the domain, allowing air to enter or exit, while the rest of the airways are assumed to be part of the porous medium. As a consequence, lung ventilation is mainly influenced by the conductivity of the porous medium. A simple approach to represent the interaction between the airway tree and lung parenchyma is to couple the lung continuum model with a dyadic resistive pipe network that follows the anatomical airway geometry [9, 61]. Finally, applying our model to lung diseases in which pulmonary surfactant has a critical pathophysiological role remains a future challenge. Future contributions should be analyzing the effect of an impaired and decreased pulmonary surfactant production in the lung response, which is especially critical in patients with acute respiratory distress syndrome connected to mechanical ventilation, who could present abnormally high levels of surface tension and, consequently, collapsed lung regions and highly heterogenous regional ventilation [2, 40, 64].

Acknowledgements

This work was funded by the National Agency for Research and Development (ANID) of Chile through the grant FONDECYT Regular # 1220465. NA-R acknowledges the support of the graduate fellowship ANID BECAS/DOCTORADO NACIONAL 21212320.

References

- [1] Albaiceta, G.M., Blanch, L., Lucangelo, U., 2008. Static pressure–volume curves of the respiratory system: were they just a passing fad? *Current opinion in critical care* 14, 80–86.
- [2] Albert, R.K., 2012. The role of ventilation-induced surfactant dysfunction and atelectasis in causing acute respiratory distress syndrome. *American journal of respiratory and critical care medicine* 185, 702–708.
- [3] Avery, M.E., Mead, J., 1959. Surface properties in relation to atelectasis and hyaline membrane disease. *AMA journal of diseases of children* 97, 517–523.
- [4] Avilés-Rojas, N., Hurtado, D.E., 2022. Whole-lung finite-element models for mechanical ventilation and respiratory research applications. *Frontiers in Physiology* , 2060.
- [5] Bachofen, H., Hildebrandt, J., Bachofen, M., 1970. Pressure-volume curves of air-and liquid-filled excised lungs-surface tension in situ. *Journal of applied physiology* 29, 422–431.
- [6] Ball, L., Dameri, M., Pelosi, P., 2015. Modes of mechanical ventilation for the operating room. *Best Practice & Research Clinical Anaesthesiology* 29, 285–299.
- [7] Bates, J.H., 2009. *Lung Mechanics. An Inverse Modeling Approach*.
- [8] Ben-Tal, A., 2006. Simplified models for gas exchange in the human lungs. *Journal of theoretical biology* 238, 474–495.
- [9] Berger, L., Bordas, R., Burrowes, K., Grau, V., Tavener, S., Kay, D., 2016. A poroelastic model coupled to a fluid network with applications in lung modelling. *International journal for numerical methods in biomedical engineering* 32.
- [10] Birzle, A.M., Wall, W.A., 2019. A viscoelastic nonlinear compressible material model of lung parenchyma—experiments and numerical identification. *Journal of the mechanical behavior of biomedical materials* 94, 164–175.
- [11] Butt, H.J., Graf, K., Kappl, M., 2023. *Physics and chemistry of interfaces*. John Wiley & Sons.
- [12] Chiew, Y.S., Tan, C.P., Chase, J.G., Chiew, Y.W., Desai, T., Ralib, A.M., Nor, M.B.M., 2018. Assessing mechanical ventilation asynchrony through iterative airway pressure reconstruction. *Computer methods and programs in biomedicine* 157, 217–224.
- [13] Choo, J., 2018. Large deformation poromechanics with local mass conservation: An enriched galerkin finite element framework. *International Journal for Numerical Methods in Engineering* 116, 66–90.
- [14] Coleman, B.D., 1964. Thermodynamics of materials with memory. *Arch. Rational Mech. Anal* 17, 1–46.
- [15] Coleman, B.D., Gurtin, M.E., 1967. Thermodynamics with internal state variables. *The journal of chemical physics* 47, 597–613.
- [16] Concha, F., Hurtado, D.E., 2020. Upscaling the poroelastic behavior of the lung parenchyma: A finite-deformation micromechanical model. *Journal of the Mechanics and Physics of Solids* 145, 104147.
- [17] Concha, F., Sarabia-Vallejos, M., Hurtado, D.E., 2018. Micromechanical model of lung parenchyma hyperelasticity. *Journal of the Mechanics and Physics of Solids* 112, 126–144.
- [18] Coussy, O., 2004. *Poromechanics*. John Wiley & Sons.
- [19] Denny, E., Schroter, R., 2000. Viscoelastic behavior of a lung alveolar duct model. *J. Biomech. Eng.* 122, 143–151.
- [20] Escolar Castellón, J.d.D., et al., 2004. Lung hysteresis: a morphological view. *Histology and histopathology* .
- [21] Fung, Y.c.B., Patitucci, P., Tong, P., 1978. Stress and strain in the lung. *Journal of the Engineering Mechanics Division* 104, 201–223.
- [22] Gibson, G., Pride, N., 1976. Lung distensibility. the static pressure-volume curve of the lungs and its use in clinical assessment. *British journal of diseases of the chest* 70, 143–184.
- [23] Goerke, J., 1998. Pulmonary surfactant: functions and molecular composition. *Biochimica et Biophysica Acta (BBA)-Molecular Basis of Disease* 1408, 79–89.
- [24] Griesse, M., 1999. Pulmonary surfactant in health and human lung diseases: state of the art. *European Respiratory Journal* 13, 1455–1476.
- [25] Harris, R.S., 2005. Pressure-volume curves of the respiratory system. *Respiratory care* 50, 78–99.

- [26] Hess, D.R., 2014. Respiratory mechanics in mechanically ventilated patients. *Respiratory Care* 59, 1773–1794.
- [27] Holzapfel, G.A., 2002. Nonlinear solid mechanics: a continuum approach for engineering science.
- [28] Hubmayr, R.D., 2002. Perspective on lung injury and recruitment: a skeptical look at the opening and collapse story. *American journal of respiratory and critical care medicine* 165, 1647–1653.
- [29] Hurtado, D.E., Avilés-Rojas, N., Concha, F., 2023. Multiscale modeling of lung mechanics: From alveolar microstructure to pulmonary function. *Journal of the Mechanics and Physics of Solids* , 105364.
- [30] Hurtado, D.E., Castro, S., Madrid, P., 2017a. Uncertainty quantification of 2 models of cardiac electromechanics. *International journal for numerical methods in biomedical engineering* 33, e2894.
- [31] Hurtado, D.E., Villarroel, N., Andrade, C., Retamal, J., Bugedo, G., Bruhn, A.R., 2017b. Spatial patterns and frequency distributions of regional deformation in the healthy human lung. *Biomechanics and Modeling in Mechanobiology* 16, 1413–1423.
- [32] Hurtado, D.E., Zavala, P., 2021. Accelerating cardiac and vessel mechanics simulations : An energy-transform variational formulation for soft-tissue hyperelasticity. *Computer Methods in Applied Mechanics and Engineering* 379, 113764.
- [33] Ingenito, E.P., Tsai, L.W., Majumdar, A., Suki, B., 2005. On the role of surface tension in the pathophysiology of emphysema. *American journal of respiratory and critical care medicine* 171, 300–304.
- [34] Ismail, M., Comerford, A., Wall, W., 2013. Coupled and reduced dimensional modeling of respiratory mechanics during spontaneous breathing. *International journal for numerical methods in biomedical engineering* 29, 1285–1305.
- [35] Jbaily, A., Frank, S., Szeri, A.J., 2020. Pulmonary mechanics and gas exchange: a mathematical framework. *International Journal of Engineering Science* 154, 103276.
- [36] Jonson, B., Svantesson, C., 1999. Elastic pressure–volume curves: what information do they convey? *Thorax* 54, 82–87.
- [37] Kralchevsky, P.A., Danov, K.D., Denkov, N.D., 1997. Chemical physics of colloid systems and interfaces. *Handbook of surface and colloid chemistry* 2.
- [38] Kylstra, J.A., Schoenfisch, W.H., 1972. Alveolar surface tension in fluorocarbon-filled lungs. *Journal of Applied Physiology* 33, 32–35.
- [39] Lange, N.R., Schuster, D.P., 1999. The measurement of lung water. *Critical Care* 3, 19–24.
- [40] Levy, M.M., 2002. Optimal peep in ards: Changing concepts and current controversies. *Critical care clinics* 18, 15–33.
- [41] Li, C., Borja, R.I., Regueiro, R.A., 2004. Dynamics of porous media at finite strain. *Computer methods in applied mechanics and engineering* 193, 3837–3870.
- [42] Logg, A., Mardal, K.A., Wells, G.N., et al., 2012. *Automated Solution of Differential Equations by the Finite Element Method*. Springer.
- [43] Lu, N., Zhang, C., 2019. Soil sorptive potential: Concept, theory, and verification. *Journal of Geotechnical and Geoenvironmental Engineering* 145, 04019006.
- [44] Lubliner, J., 1973. On the structure of the rate equations of materials with internal variables. *Acta Mechanica* 17, 109–119.
- [45] Ma, B., Breen, B., Bates, J.H., 2013. Influence of parenchymal heterogeneity on airway-parenchymal interdependence. *Respiratory physiology & neurobiology* 188, 94–101.
- [46] Ma, H., Fujioka, H., Halpern, D., Bates, J.H., Gaver III, D.P., 2023. Full-lung simulations of mechanically ventilated lungs incorporating recruitment/derecruitment dynamics. *Frontiers in Network Physiology* 3.
- [47] Ma, H., Fujioka, H., Halpern, D., Gaver III, D.P., 2020. Surfactant-mediated airway and acinar interactions in a multi-scale model of a healthy lung. *Frontiers in Physiology* 11, 941.
- [48] MacMinn, C.W., Dufresne, E.R., Wettlaufer, J.S., 2016. Large deformations of a soft porous material. *Physical Review Applied* 5, 044020.
- [49] Manikantan, H., Squires, T.M., 2020. Surfactant dynamics: hidden variables controlling fluid flows.

- Journal of fluid mechanics 892, P1.
- [50] Matamis, D., Lemaire, F., Harf, A., Brun-Buisson, C., Ansquer, J., Atlan, G., 1984. Total respiratory pressure-volume curves in the adult respiratory distress syndrome. *Chest* 86, 58–66.
 - [51] Maury, B., 2013. *The Respiratory System in Equations*. Springer-Verlag Italia.
 - [52] Mead, J., Whittenberger, J., Radford Jr, E., 1957. Surface tension as a factor in pulmonary volume-pressure hysteresis. *Journal of applied physiology* 10, 191–196.
 - [53] Middleton, S., Dimbath, E., Pant, A., George, S.M., Maddipati, V., Peach, M.S., Yang, K., Ju, A.W., Vahdati, A., 2022. Towards a multi-scale computer modeling workflow for simulation of pulmonary ventilation in advanced covid-19. *Computers in Biology and Medicine* 145, 105513.
 - [54] Morris, J., Ingenito, E., Mark, L., Kamm, R., Johnson, M., 2001. Dynamic behavior of lung surfactant. *J. Biomech. Eng.* 123, 106–113.
 - [55] Morton, S.E., Dickson, J., Chase, J.G., Docherty, P., Desai, T., Howe, S.L., Shaw, G.M., Tawhai, M., 2018. A virtual patient model for mechanical ventilation. *Computer Methods and Programs in Biomedicine* 165, 77–87.
 - [56] v. Neergaard, K., 1929. Neue auffassungen über einen grundbegriff der atemmechanik: Die retraktionskraft der lunge, abhängig von der oberflächenspannung in den alveolen. *Zeitschrift für die gesamte experimentelle Medizin* 66, 373–394.
 - [57] Nikoee, E., Habibagahi, G., Hassanizadeh, S.M., Ghahramani, A., 2013. Effective stress in unsaturated soils: A thermodynamic approach based on the interfacial energy and hydromechanical coupling. *Transport in porous media* 96, 369–396.
 - [58] Otis, D.R., Ingenito, E.P., Kamm, R.D., Johnson, M., 1994. Dynamic Surface-Tension of Surfactant Ta - Experiments and Theory. *Journal of Applied Physiology* 77, 2681–2688.
 - [59] Patte, C., Genet, M., Chapelle, D., 2022. A quasi-static poromechanical model of the lungs. *Biomechanics and Modeling in Mechanobiology* 21, 527–551.
 - [60] Pattle, R., 1955. Properties, function and origin of the alveolar lining layer. *Nature* 175, 1125–1126.
 - [61] Pozin, N., Montesantos, S., Katz, I., Pichelin, M., Vignon-Clementel, I., Grandmont, C., 2017. A tree-parenchyma coupled model for lung ventilation simulation. *International journal for numerical methods in biomedical engineering* 33, e2873.
 - [62] Quiros, K., Nelson, T., Sattari, S., Mariano, C., Ulu, A., Dominguez, E., Nordgren, T., Eskandari, M., 2022. Mouse lung mechanical properties under varying inflation volumes and cycling frequencies. *Scientific reports* 12, 7094.
 - [63] Raghavendran, K., Willson, D., Notter, R., 2011. Surfactant therapy for acute lung injury and acute respiratory distress syndrome. *Critical care clinics* 27, 525–559.
 - [64] Retamal, J., Hurtado, D.E., Villarroel, N., Bruhn, A., Bugeo, G., Amato, M.B.P., Costa, E.L.V., Hedenstierna, G., Larsson, A., Borges, J.B., 2018. Does Regional Lung Strain Correlate With Regional Inflammation in Acute Respiratory Distress Syndrome During Nonprotective Ventilation? An Experimental Porcine Study. *Critical Care Medicine* 46, 591–599.
 - [65] Roth, C.J., Ismail, M., Yoshihara, L., Wall, W.A., 2017. A comprehensive computational human lung model incorporating inter-acinar dependencies: Application to spontaneous breathing and mechanical ventilation. *International journal for numerical methods in biomedical engineering* 33, e02787.
 - [66] Scarpelli, E.M., 1977. The surfactant system of the lung. *International Anesthesiology Clinics* 15, 19–60.
 - [67] Schürch, S., Bachofen, H., Goerke, J., Green, F., 1992. Surface properties of rat pulmonary surfactant studied with the captive bubble method: adsorption, hysteresis, stability. *Biochimica et Biophysica Acta (BBA)-Biomembranes* 1103, 127–136.
 - [68] Smith, J.C., Stamenovic, D., 1986. Surface forces in lungs. i. alveolar surface tension-lung volume relationships. *Journal of applied physiology* 60, 1341–1350.
 - [69] Steimle, K.L., Mogensen, M.L., Karbing, D.S., de la Serna, J.B., Andreassen, S., 2011. A model of ventilation of the healthy human lung. *computer methods and programs in biomedicine* 101, 144–155.
 - [70] Suki, B., Stamenovic, D., Hubmayr, R., 2011. Lung parenchymal mechanics. *Comprehensive Physiology* 1, 1317.

- [71] Sun, W., Ostien, J.T., Salinger, A.G., 2013. A stabilized assumed deformation gradient finite element formulation for strongly coupled poromechanical simulations at finite strain. *International Journal for Numerical and Analytical Methods in Geomechanics* 37, 2755–2788.
- [72] Suter, P.M., Fairley, H.B., Isenberg, M.D., 1975. Optimum end-expiratory airway pressure in patients with acute pulmonary failure. *New England Journal of Medicine* 292, 284–289.
- [73] Takeuchi, M., Sedeek, K.A., Schettino, G.P., Suchodolski, K., Kacmarek, R.M., 2001. Peak pressure during volume history and pressure–volume curve measurement affects analysis. *American journal of respiratory and critical care medicine* 164, 1225–1230.
- [74] Tawhai, M.H., Nash, M.P., Lin, C.L., Hoffman, E.A., 2009. Supine and prone differences in regional lung density and pleural pressure gradients in the human lung with constant shape. *Journal of Applied Physiology* 107, 912–920.
- [75] The CGAL Project, 2021. CGAL User and Reference Manual. 5.3.1 ed., CGAL Editorial Board. URL: <https://doc.cgal.org/5.3.1/Manual/packages.html>.
- [76] Vandamme, M., Brochard, L., Lecampion, B., Coussy, O., 2010. Adsorption and strain: the co₂-induced swelling of coal. *Journal of the Mechanics and Physics of Solids* 58, 1489–1505.
- [77] Veldhuizen, R., Nag, K., Orgeig, S., Possmayer, F., 1998. The role of lipids in pulmonary surfactant. *Biochimica et Biophysica Acta (BBA)-Molecular Basis of Disease* 1408, 90–108.
- [78] Venegas, J.G., Harris, R.S., Simon, B.A., 1998. A comprehensive equation for the pulmonary pressure–volume curve. *Journal of applied physiology* 84, 389–395.
- [79] Vuong, A.T., Yoshihara, L., Wall, W., 2015. A general approach for modeling interacting flow through porous media under finite deformations. *Computer Methods in Applied Mechanics and Engineering* 283, 1240–1259.
- [80] West, J.B., 2012. *Respiratory Physiology: The Essentials*. Ninth edition ed., Lippincott Williams & Wilkins.
- [81] Wilson, T.A., 1982. Surface tension–surface area curves calculated from pressure–volume loops. *Journal of Applied Physiology* 53, 1512–1520.
- [82] Zhang, Y., 2018. Mechanics of adsorption–deformation coupling in porous media. *Journal of the Mechanics and Physics of Solids* 114, 31–54.
- [83] Zhao, Q., Ren, W., Zhang, Z., 2021. A thermodynamically consistent model and its conservative numerical approximation for moving contact lines with soluble surfactants. *Computer Methods in Applied Mechanics and Engineering* 385, 114033.
- [84] Zhou, C., Chase, J.G., Knopp, J., Sun, Q., Tawhai, M., Möller, K., Heines, S.J., Bergmans, D.C., Shaw, G.M., Desai, T., 2021. Virtual patients for mechanical ventilation in the intensive care unit. *Computer Methods and Programs in Biomedicine* 199, 105912.